



Amazon Keyspaces

Lecture Slides

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Good luck on your exam!

Amazon Keyspaces



Amazon Keyspaces is a fully managed Apache Cassandra database. Cassandra is an open-source NoSQL key/value database similar to DynamoDB in that is columnar store database but has some additional functionality. *When you want to use Apache Casandra.*



Cluster — a collection of nodes

Nodes — holds 2 – 4 TB of data

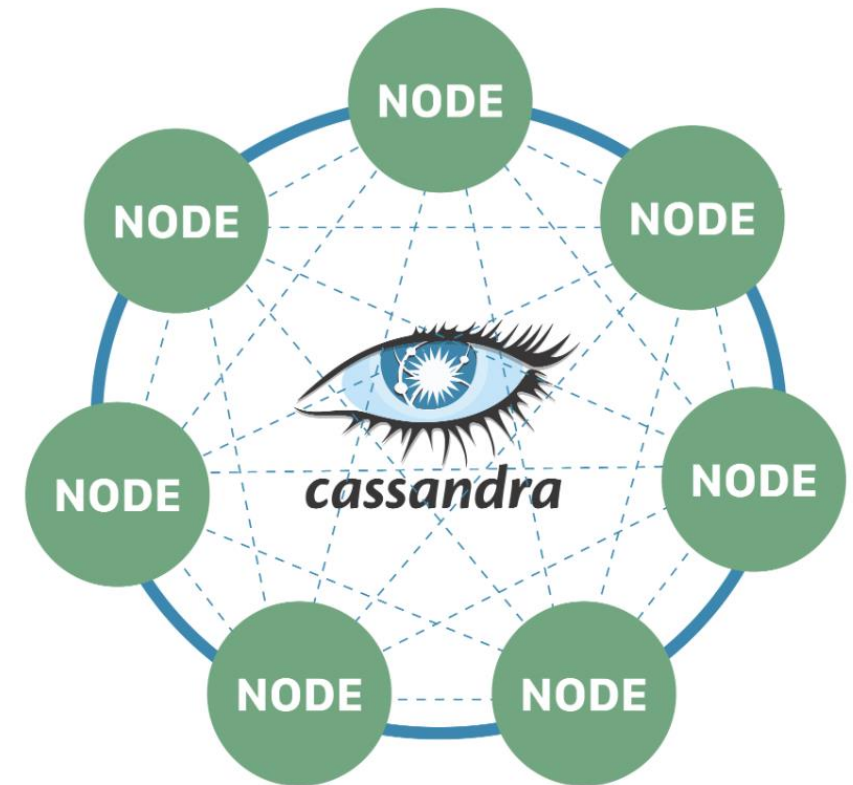
- All nodes read and write
- Nodes represent smallest unit of a database
- Data is replicated on multiple nodes

Ring — Nodes are arranged in a ring where all nodes connect to each other

Keyspace — a namespace that specifies data replication on nodes

Table — tabular data of columns and rows with a primary key

- Casandra is queried using the Casandra Query Language (CQL)
- Interacting with Casandra is typically performed via an SDK in your programming language of choice
- Amazon Keyspaces allows you to perform the following the AWS Management Console:
 - Create Keyspaces
 - Create Tables
 - CQL editor (query via the UI)



Cassandra Query Language

Cassandra Query Language (CQL) is a language similar to SQL and the primary language to interact with a Cassandra database

Create Keyspace

```
CREATE KEYSPACE IF NOT EXISTS exampleKeyspace  
WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 3};
```

Create Table

```
CREATE TABLE IF NOT EXISTS exampleKeyspace.users (  
    user_id uuid PRIMARY KEY,  
    first_name text,  
    last_name text,  
    email text  
);
```

Insert Data

```
INSERT INTO exampleKeyspace.users (user_id, first_name, last_name, email)  
VALUES (uuid(), 'John', 'Doe', 'johndoe@example.com');
```

Query Data

```
SELECT *  
FROM exampleKeyspace.users  
WHERE user_id = 12345678-1234-5678-1234-567812345678;
```

Update Data

```
UPDATE exampleKeyspace.users  
SET first_name = 'Jane'  
WHERE user_id = 12345678-1234-5678-1234-567812345678;
```

Delete Data

```
DELETE FROM exampleKeyspace.users  
WHERE user_id = 12345678-1234-5678-1234-567812345678;
```